

PCBs in the Hudson: Pollution, Contamination, and Billions Spent

An Analysis of the History of Regulation, Litigation, and Methods Surrounding General Electric's Dumping of PCBs in the Hudson River

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A General Electric dredging barge lifts soil from a Hudson River bank near Troy, N.Y., in May 2015, as part of an effort to remove PCBs (Brooks 2016).

1 Introduction and Background

I grew up two blocks south of Washington Square Park, in lower Manhattan. Starting in fifth grade, I went to the United Nations International School (UNIS) located on 24th Street and the East River. Nearly every single day on my walk, bike, or bus to school, I would pass the massive ConEd power generation station. I remember sitting in 6th grade science and hearing, for the first time, about the ConEd power plant. We talked about how the amazing city we all called home was able to function. We discussed how we all needed power to heat our houses and run our buildings, and that the ConEd plant was an example of just that.

In high school, I took the International Baccalaureate Higher Level Chemistry exam in which we had to study methods of energy production and their sideeffects. While studying for the exam, my classmates and I researched the ConEd power plant and its runoff into the East River, and the estuary of the Hudson River (Hogan 2017). I used this knowledge as a starting point for this paper. While researching, I discovered that a much more litigated and debated topic is the larger concept of power plants dumping into the Hudson River as a whole. During my research, I discovered extensive literature and litigation detailing General Electric's history of dumping and contaminating the Hudson River upstream of New York City, in the Upper Hudson area of New York State (NOAA 2013).

The 2013 National Oceanic and Atmospheric Administration (NOAA) report detailing the contamination of the Hudson River by General Electric examines data from 2002 to 2008 from two plants, one located in Hudson Falls and the other in Fort Edward, NY. The report focuses on the polychlorinated biphenyls (PCBs) released from the facilities. The NOAA report concludes these two plants are the predominant historical source of the PCB contamination of the Hudson River ecosystem (NOAA 2013). The Environmental Protection Agency (EPA) estimates that between the 1940s and 1977, the General Electric plants released up to 1,300,000 pounds of PCBs in the Hudson River ecosystem (NOAA 2013). According to the report, even after direct discharges stopped in 1977, the facilities continued to release PCBs through groundwater from the bedrock beneath the plants, which

eventually led to erosion of contaminated riverbanks (NOAA 2013).

The primary takeaway from the NOAA report was that the Hudson River ecosystem remains extensively and dangerously contaminated with PCBs both directly and indirectly released from General Electric facilities. This dumping created a “serious and long-term threat” to both environmental and human health (NOAA 2013, Page 23). The report highlights that decades later the contamination still remains and spans more than 200 miles downstream, reaching the New York Harbor. Over 80 percent of the samples tested since 1975 contained PCB levels 10 to 10,000 times higher than the EPA sets as safe limits for wildlife and human consumption (NOAA 2013, Page 11).

Furthermore, the NOAA report concluded that surface sediment concentrations of PCBs were well in excess of the levels associated with toxic effects on organisms that are native to the riverbed. PCB levels were detected as high as 1650 parts per million (ppm) in topsoil sediments—which is defined as the top 12 inches of soil (NOAA 2013, Page 13). The NOAA paper cites the New York State Department of Environmental Conservation (NYSDEC) 1999 standard that leaves exceeding 0.042 ppm for wildlife result in wildlife bioaccumulation, levels above 0.58 ppm result in chronic benthic effect, and levels above 83 ppm for acute benthic effects. The 1650 ppm found in the topsoil sediment samples are therefore significantly exceeding levels that “are considered to be contaminated, and [are] potentially causing harmful impact” (NOAA 2013, Page 13).

Additionally, PCB levels in edible fish portions have consistently exceeded the United States Food and Drug Administration’s (FDAs) tolerance levels, with some samples that were collected since 2000 exceeding the human health objective. The FDA does not allow the sale of commercial fish for human consumption if the PCB levels exceed 2 ppm. In the fish samples collected, 75 to 90 percent of largemouth bass, smallmouth bass, brown bullhead, and yellow perch filets exceed this 2 ppm maximum (NOAA 2013, Page 16). In fact, 40 to 70 percent of the samples exceeded the level by a factor of 10 or more (NOAA 2013, Page 16).

Most importantly, the NOAA report demonstrates the PCBS have bioaccumulated at every level of the aquatic and terrestrial food chain. Levels found in mink and birds are greater than levels associated with reproductive impairment. Levels found in snapping turtles exceed levels associated with death risk in juvenile populations (NOAA 2013, Page 3). The case study of the tree swallows in the area is particularly horrific. Tree swallows are a clear case for bioaccumulation of PCBs, as they are an animal that lives near river beds and eats insects that traditionally inhabit river bottoms. In the mid-1990s, young juvenile tree swallow samples were found to contain PCBs at a concentration up to 62 ppm. These are “among the highest PCB levels ever reported in tree swallows from any location” (NOAA 2013, Page 24).

The NOAA report concludes with the statement that the data “suggest that PCBs are likely to be causing serious adverse effects to the area’s biota” (NOAA 2013, Page 3). This report was published in January of 2013. In February of 2014, NOAA published a blog post entitled “NOAA, U.S. Fish and Wildlife Service Correct GE’s Misinformation in Latest Hudson River Pollution Report” in which they use the data from the 2013 report to explicitly explain how General Electric has hurt the Hudson Valley and is spreading misinformation on the subject (NOAA 2014). Less than half a year later, in June and July of 2014, General Electric settled two large disputes surrounding its role in contaminating the Hudson River. A \$7.95 million lawsuit settlement with some of the Saratoga County Towns and a \$24 million EPA enforcement settlement (Times Union 2014; U.S. Environmental Protection Agency 2014).

Two years after these settlements, in the fall of 2016, I started attending UNIS and travelling past the ConEd power plant on my way to school. ConEd has yet to be convicted in a lawsuit of directly polluting the Hudson River. However, there is a current ongoing lawsuit in which the City Club of New York is suing ConEd for allegedly illegally dumping toxic wastewater containing chloroform, selenium, lead, chromium, and copper into the Hudson River at Pier 98 for the last 20 years. As of August 2026, no agreement has been reached

(W42ST 2023; ABC News 2022).

2 History of Regulation and Litigation

There is a long history of prolonged and complicated litigation tied to General Electric and the Hudson River. Understanding this history is essential to making sense of the settlements finally reached in June and July of 2014.

In 1984, the EPA, which had the Hudson River listed on its National Priorities List, issued a formal Record of Decision (ROD) stating that “no action” was needed to prevent PCB contamination in riverbeds and fish (U.S. Environmental Protection Agency 2002). However, this decision was based on outdated legal standards. Five days before the EPA signed the ROD, the FDA’s new, stricter standard of 2 ppm, as opposed to 5 ppm, for PCB contamination in fish flesh came into effect. The EPA’s ROD of “no action” was based on the older, outdated figure of 5 ppm (Hudson River Sloop Clearwater 2001). In 1989, the NYS-DEC petitioned the EPA to reconsider the 1984 “no action” decision, citing continued high and unsafe levels of PCBs in Hudson River fish. In a response document, the EPA agreed that developments since 1984 made it appropriate to undertake a more comprehensive reassessment of the “no action” decision (U.S. Environmental Protection Agency 1989). These developments included technological advances in the treatment and destruction of PCBs, which made more permanent remedies feasible to consider (U.S. Environmental Protection Agency 1989).

Between 1989 and 2000, the EPA conducted a detailed Reassessment and Remedial Investigation and Feasibility Study (RI/FS). This two-phase, 11-year project attempted to use technical advances in sediment dredging and treatment to reassess the status of PCB contamination in the Hudson River. In July of 1998, the EPA released a report on Low Resolution Sediment Coring, which found that the 1984 ROD’s assumption of declining PCB levels was not only incorrect, but that, in actuality, PCBs were mobilizing throughout the river (U.S.

Environmental Protection Agency 1998). These findings led to a new EPA ROD in February of 2002, which mandated massive dredging of contaminated sediment and riverbeds. The new ROD called for the removal of 2.65 million cubic yards of PCB-contaminated sediment from the Upper Hudson River, a complete reversal of the “no action” decision announced in 1984 (U.S. Environmental Protection Agency 2002). General Electric was designated the responsible party in the ROD and required to cover the cost (U.S. Environmental Protection Agency 2002). In October 2005, General Electric officially signed a Consent Decree (CD) with the EPA in which it agreed to perform the first phase of the environmental dredging and construct the sediment-processing facility (U.S. Department of Justice 2006). In May of 2006, the United States District Court in Albany formally approved the CD, making it legally enforceable (U.S. Environmental Protection Agency 2006).

In 2006, the Natural Resources Defense Council (NRDC) filed a lawsuit against the EPA demanding release of critical documents on the cleanup plan, alleging the EPA was weakening environmental standards (Natural Resources Defense Council 2006). This lawsuit was running parallel to the negotiation between the EPA and General Electric during debates surrounding the CD. After the ratification of the CD in May of 2006, the NRDC continued to criticize the EPA. The NRDC claimed that the EPA caved to General Electric, as they retained the ability to opt out of the second phase of the cleanup if they reimbursed the EPA for the cost (Cho 2010). In December 2010, however, General Electric notified the EPA that it would conduct and fund the second phase of the cleanup operation (Cho 2010). From 2009 to 2015, phases one and two of the cleanup project took place, costing General Electric 1.7 billion dollars (Fox Business 2017).¹ In total, 2.75 million cubic yards of contaminated sediment were removed, and over 310,000 lbs of PBCs were excavated (U.S. Environmental Protection Agency 2017; U.S. Environmental Protection Agency 2026).

In 2009, Saratoga County towns filed a federal lawsuit surrounding unexpected conse-

¹As of 2015, the EPA sights the cost of the clean up effort at \$1.5 billion (U.S. Environmental Protection Agency 2026). The Fox Business value of \$1.7 billion is from 2017 and likely accounts for additional cost in those final two years.

quences of the massive dredging process. The towns of Halfmoon, Stillwater, and Waterford sued General Electric for water supply disruptions caused by the dredging operations. During the dredging mandated by the aforementioned CD with the EPA, PCB concentrations in the towns’ water spiked, forcing the towns to shut down or relocate water intakes at a cost of millions of dollars (Times Union 2014). During the discovery phase of the lawsuit, more than ten million pages of General Electric’s internal records were released, including extensive documentation of the company’s knowledge of PCB contamination and continued operations (Times Union 2014). In July of 2014, General Electric settled with the Towns of Stillwater and Waterford for \$7.95 million (Times Union 2014).² In May of 2016, the Saratoga County Water Authority settled for an additional \$5.3 million (Times Union 2016).

3 Methods Review

The following section of this paper will focus on the methods used to conduct the research that caused the countless regulations and lawsuits discussed above. I will start by analyzing the methods of Butcher, Gauthier, and Garvey’s 1997 paper, the “Use of Historical PCB Aroclor Measurements: Hudson River Fish Data”, to understand the scientific data and methods used by the EPA.³ I will then discuss Field, Kern, and Rosman’s 2016 paper “Re-Visiting Projections of PCBs in Lower Hudson River Fish Using Model Emulation” to have a more modern methodology. Comparing the two papers from nearly 20 years apart will contextualize how and if the methods for looking for large-scale PCB contamination have changed.

Butcher, Gauthier, and Garvey’s 1997 paper, “Use of Historical PCB Aroclor Measurements: Hudson River Fish Data”, focuses on the standardization of historical data across changing analytical technologies. Butcher, Gauthier, and Garvey were attempting to rec-

²The Town of Halfmoon settled for a non specified amount in July of 2016 (Times Union 2016).

³Butcher, Gauthier, and Garvey 1997 is cited in the EPA’s Second Five-Year Review Report for the Hudson River PCBs Superfund Site from 2017 as an example of historical methodology (U.S. Environmental Protection Agency 2017).

oncile historical methodologies to bridge the gap between older and newer analytical techniques. The paper used more than 20 years of historical fish tissue data, beginning in 1977, based on packed-column gas chromatography (GC), which reported PCBs as “Aroclor equivalents”, and compared it to 1993 samples analyzed with more modern capillary-column GC (Butcher, Gauthier, and Garvey 1997).⁴⁵ The authors developed mathematical models to estimate what the older packed-column methods would have measured if they had been applied to the newer congener-specific data. They then applied linear regressions to correct the older “Aroclor” quantitation schemes used from 1977 to 1982 to match the more consistent basis adopted in 1983, allowing the entire historical record of PCB contamination data to be compared on a single scale (Butcher, Gauthier, and Garvey 1997).⁶

The older packed-column GC had strong analytical limitations, as it was unable to resolve individual PCB congeners and relied on a small set of chromatogram peaks, often leading to false signals of decline as laboratory quantitation schemes changed over time. Once this comparison method was utilized, it became clear that the reported decline in PCB contamination had been overstated. The paper found that changing the quantitation peaks alone could account for a 40% artificial decline in reported Aroclor 1016 levels, making these corrections pivotal for accurate interpretations of contamination (Butcher, Gauthier, and Garvey 1997).

The Field, Kern, and Rosman paper moves away from data standardization to predictive model validation using model emulation. This technique has researchers condense complex, highly parameterized mechanistic models into computationally efficient equations using non-linear and linear regressions. The model’s estimates revealed that sediment PCB concentrations were significantly higher and more widespread than previously thought. Instead of

⁴The capillary-column GC is the more modern method as it allows for congener-specific resolution (Butcher, Gauthier, and Garvey 1997).

⁵This 1993 data is data that was collected to aid in the reevaluation of the “no action” decision made by the EPA in 1984 (Butcher, Gauthier, and Garvey 1997).

⁶The majority of the Butcher, Gauthier, and Garvey 1997 paper is figures that explain the equations created and the regressions utilized. Please refer to the paper for exact figures (Butcher, Gauthier, and Garvey 1997).

the 8% annual recovery rate predicted by the original models, the paper calculated a new recovery rate of 1.3 to 3% by comparing 1991 survey data with 2003 data (Field, Kern, and Rosman 2016).⁷

Additionally, this new predictive model showed that older models were biased, as they underestimated the initial sediment contamination levels and overstated the natural recovery rate of the area. When General Electric was forced to sample the Hudson River in the cleanup effort, it took physical samples from over 8,000 locations. This vast amount of sediment data clearly illustrated how inaccurately the older models had predicted initial sediment contamination levels. Furthermore, the original models assumed the river would naturally bury the toxic PCBs under clean layers of mud at the rapid rate of 8% a year. However, when the researchers compared the actual sediment data from 1991 to 2003, they found the true recovery rate was between 1.3% and 3% per year. This is a dramatic miscalculation, as at 8% it would only take nine years to reduce the PCB contamination by half, whereas at the 1.3% rate the reduction by half would take over 50 years (Field, Kern, and Rosman 2016).

Overall, these two papers and their approaches allow for the accurate comparison of decades of data, making the scale of the contamination and the length of the recovery period extremely apparent. Even at the cost of \$1.7 billion, plus countless additional dollars spent on litigation, PCB contamination in the Hudson River will not be reduced by half for more than 50 years (Fox Business 2017; Field, Kern, and Rosman 2016).

4 Critical Analysis

The Field, Kern, and Rosman paper is an incredible example of reevaluating past claims and retesting their validity with new tools and data. Only once the 8,000 General Electric field samples were taken were the researchers able to understand the basis of previous models

⁷The paper refers to the new annual recovery rate as the exponential decay rate. The paper treated the exponential decay rate as an empirical measure of the actual natural recovery rate of the river's sediment (Field, Kern, and Rosman 2016).

and correct them to give a more accurate recovery rate prediction (Field, Kern, and Rosman 2016).

Together, these two papers illuminate a fundamental tension in scientific knowledge and environmental regulation: analytical standards and data collection methods are constantly evolving, making historical comparisons and lengthy litigation battles inherently problematic. The combination of Butcher, Gauthier, and Garvey's equation for comparing measurements across different types of GC and Field, Kern, and Rosman's more accurate predictive model, which incorporates new sampling data, allows scientists to avoid starting from scratch with each new technical development or discovered bias (Field, Kern, and Rosman 2016). Instead, with these methods, regulators and scientists alike have the ability to examine historical and modern data to address the current and future state of PCB contamination in not just the Hudson River, but anywhere historical data and enough information to build the predictive model exist.

The largest issue within this contamination event has nothing to do with methodology, but rather with timing. Every methodological correction has arrived years after the political or regulatory decision point it should have informed. The Butcher, Gauthier, and Garvey paper was released in 1997, years after the "no action" framing had already shaped policy and the EPA's stance for over a decade (Butcher, Gauthier, and Garvey 1997). Field, Kern, and Rosman's correction came in 2016, not only after the dredging design was locked in around the older recovery assumptions but also after \$1.7 billion had already been spent on an irreversible cleanup effort (Fox Business 2017; Field, Kern, and Rosman 2016). This case illustrates a common pattern: scientific caution and methodological rigor, though well-meaning, combined with investigative efforts that came after harm had already occurred rather than before, ultimately slowed regulatory response. In doing so, these papers function less as a check on bad decisions than as a retrospective critique of decisions that had already caused harm.

5 Conclusion

Through decades of complex regulatory and litigation battles, General Electric was held responsible for contaminating the Hudson River with PCBs. After being compelled by the EPA to spend \$1.7 billion on remediation, the company also settled multimillion-dollar lawsuits with the Towns of Stillwater and Waterford, and later the Saratoga County Water Authority, over water-supply disruptions the dredging itself caused. Significant research by Butcher, Gauthier, and Garvey, as well as by Field, Kern, and Rosman, has enabled the evaluation of historical data across different gas chromatography methods, allowing researchers to build more accurate predictive models for recovery timelines. Based on these models, by the time my children walk past the Con Ed power plant on their way to school and ask the same question I posed in my sixth-grade science class, the Hudson River will still be contaminated with PCBs. For communities that rely on the river for fishing, this slow institutional reckoning has meant generations without safe access to a resource their community once depended on.

The most problematic continuity in this case study is not ecological but institutional: the same reluctance to commit to a hard conclusion that shaped the 1984 “no action” decision still shapes the EPA’s posture in 2026. The EPA is still negotiating the same question it faced then, whether the cleanup has actually worked, and is doing so with the same institutional reluctance to give a definitive answer. The EPA’s own third five-year review, finalized in January 2025 more than two years behind its original schedule, echoes the delays that defined this entire story: rather than answering whether the cleanup actually worked, it deferred that determination to a future addendum, contingent on more fish-tissue data, that may not arrive until as late as 2027 (U.S. Environmental Protection Agency 2024, 2025). Politically, the case did force General Electric to internalize over a billion dollars of cleanup cost it would rather not have paid, and it did establish that a corporation could be compelled, after enough litigation, to fund remediation on such a large scale. But it did not fundamentally change the EPA’s institutional posture toward contamination sites; it simply

proved that the posture could eventually be forced into partial action, given enough pressure and many decades.

In that same science class where I first learned about the Con Ed power plant and how it functioned, I also learned about restoration efforts. As a class, we researched the Billion Oyster Project, a nonprofit organization that engages one million people in efforts to restore a billion oysters to help clean New York Harbor by 2035. As of 2025, the organization has repopulated 75 million juvenile oysters in the harbor and recycled more than 1.8 million pounds of oyster shell from NYC restaurants (Billion Oyster Project 2025). Yet despite these restoration successes, contamination persists, and regulatory action remains slow. This underscores that meaningful remediation depends not solely on litigation and policy, but also on direct environmental intervention by committed actors.

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